

Browser + manifest.json
(name, icons, standalone)



Service worker (sw.js):
precache-v3 icons +
offline page



Runtime cache v4 (80
entries, LRU prune)



API: network-only +
offline JSON



Navigation: network-first
+ offline page



Statics:
stale-while-revalidate



Updates: skipWaiting +
claim + banner + version
poll